

MATTHEW CAMP

Huntsville, AL ■ 423.322.3976 ■ resume@aicampground.com

PROFILE

Strategic leader and data scientist with a strong foundation in software development, deterministic systems, and distributed architectures. Proven ability to drive high-impact initiatives in both government and private sectors, combining leadership excellence with technical expertise in NixOS, Python, and Julia. Adept at developing integrated software systems and leading high-performing teams to exceed organizational objectives through innovative solutions and operational precision.

- Demonstrated success in developing integrated software systems. Record of success in both government and private sectors.
- Respected for capacity to apply systematic approach and ability to thrive in challenging environments requiring strong critical thinking, investigative, and analytical skills.
- Recognized as the driving force behind highly successful projects that exceed organizational expectations.
- Well-versed in creating high-performance cultures and focusing teams on operational excellence.
- Current Top Secret/Specialized Compartmentalized Information (TS/SCI) with Counterintelligence Polygraph.

CORE COMPETENCIES

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|------------------------------|------------------------------------|-----------------------------------|
| • Machine Learning | • Operational Streamlining | • Risk Mitigation/Management |
| • Team Leadership/Management | • All-Source Intelligence Analysis | • Planning/Organization/Logistics |
| • Advanced Analytics | • DevOps | • Process Improvement Strategies |

PROFESSIONAL EXPERIENCE

RELEVANT EXPERIENCE

Advanced Technology Applications (ATA) LLC - Great Falls, VA (Remote)

Senior Data Scientist / Full Stack MLOps/DevOps Engineer

November 2021 - Present

As a key figure on the platform team, I spearheaded the deployment and management of intricate technological solutions. Leveraging expertise in JupyterHub, Kubernetes, Docker, Nix, and Flink, I facilitated seamless tool integration and customization to optimize workflow efficiency. Recognizing areas for improvement, I proactively mastered Nix to design robust solutions, including replacing unreliable Ansible playbooks and creating reproducible configurations for Flink jobs, enabling consistent deployments across local and Kubernetes environments. My efforts extended to building a local development cluster with K3D and Nix, allowing developers to spin up Kubernetes environments with a single command. Dedicated to mentorship and knowledge-sharing, I cultivated a collaborative team culture through daily coaching sessions for junior developers. My innovative strategies in deployment, data science, and DevSecOps have significantly enhanced organizational processes and positioned the team for diverse challenges.

Key Accomplishments

- Spearheaded the adoption of Nix and NixOS creating a secure and reproducible DevSecOps framework and addressing challenges like configuration drift and dependency conflicts.
- Developed a UNet-based image segmentation model in Julia including dataset creation with Label Studio and performance tracking with MLFlow.
- Streamlined Flink job deployment for both local and Kubernetes environments by leveraging Nix for reproducible configurations and consistent runtime environments.
- Built a local development cluster with K3D and Nix enabling developers to spin up Kubernetes environments with a single command for testing and debugging.
- Deployed JupyterHub on enterprise Kubernetes clusters customized Helm charts and developed Docker images supporting Julia Python and R workflows.
- Designed modular infrastructure solutions for systems like Kafka and MLFlow simplifying deployment improving scalability and enhancing security.

- Mentored junior associates in emerging technologies fostering team collaboration and streamlining onboarding with Nix development shells.

Colsa - Huntsville, AL

Artificial Intelligence Software Developer / Full Stack Data Scientist / Tech Lead September 2018 - November 2021

Developed data science strategies for dealing with big data problems. Primary driving force behind the adoption of modern software development practices by the government customer. Assisted junior developers become better familiar with Linux, Python and a number of DevOps tools. Interacted with customer Technical Point of Contact (TPOC) daily to ensure contract needs were being satisfied.

Key Accomplishments

- Initiated the organizational adoption of DevOps tools such as Gitlab CI/CD Pipelines and Docker
- Created a GraphQL API in Python utilizing the Graphene MongoEngine and a self-written Neo4j API.
- Wrote an unsupervised machine learning algorithm for describing data found in domain specific files.
- Architected and developed a data pipeline using a combination of Docker and Apache Airflow.
- Built and implemented a domain specific graph data (MongoDB + Neo4j) model for use in maintaining a file meta database.

Science Applications International Corporation (SAIC) - Huntsville, AL

Senior Database Administrator & Business Process Improvement Analyst August 2017 - August 2018

Sole technical developer in the security department, responsible for fulfilling data needs of senior leadership as well as being instrumental in the transition processes from a legacy Access Commander database system to a Sims database system. Rapid development of automated solutions using Python to provide a high level of integration with existing enterprise systems, greatly increasing the efficiency of human labor. Worked with subject matter experts to systematically define incomplete or non-existent business processes. Began the implementation of Lean practices in the management of department wide projects using a self-hosted web-based Kanban system.

Key Accomplishments

- Developed real-time automated reporting solutions for enterprise wide consumption using a Flask web app architecture and SQL Server
- Created Python based web scrapers with Selenium and Autoit to autonomously download reports from parts of the enterprise without readily available APIs
- Built a web-based user interface for service center personnel to be able to efficiently process hundreds of security clearance prescreens weekly
- Took the initiative to develop a nearly autonomous contract closeout system to eliminate hundreds of backlogged contracts and decrease the number of man hours required for the process going forward.

ExoAnalytic Solutions - Huntsville, AL

Algorithm Developer April 2016 - August 2017

Assisted the Chief Information Officer as the local information technology (IT) support lead for the Huntsville office. Researched, purchased, and built computer systems for deep learning. Regularly worked with large datasets of over several million records. Quickly adapted to fill many miscellaneous technical and non-technical requirements of a small office

Key Accomplishments

- Investigated cutting edge deep learning algorithms to include Recurrent Neural Networks Convolutional Neural Networks and Generative Adversarial Networks to aid in space situational awareness (SSA) and missile defense applications using Python and Tensorflow.
- Using a combination of both R and Python developed machine learning models based on EO/IR sensor data for both the Missile Defense and space situational awareness domains.
- Orchestrated the conversion of Missile Defense data from complicated MATLAB structures to a faster and more efficient SQLite database.

- Launched an effort to convert the company's Customer Relationship Management (CRM) system to a graph database using Neo4j software.
- Used Python to build automated web scrapers for mining data from multiple online sources. Provided increased situational awareness of the local business environment.

Elbit Systems of America - Huntsville, AL

Marketing Intern

September 2015 - April 2016

Assisted with product demonstrations for prospective customers as needed. Built PowerPoint presentations to be briefed to the CEO and other executive level leaders. Researched and analyzed product details and specifications to synthesize a simplified fact-sheet for the many aerospace offerings of the company. Initiated the development of a Customer Relationship Management system to be used in house by the business development associates.

Key Accomplishments

- Engaged with business development associates to devise requirements for a Customer Relationship Management (CRM) system.
- Architected the database schema for a CRM tool.
- Researched the federal acquisition process and utilized GovWin software to analyze key stakeholders.

Intelligence Support Battalion, FHG, Marine Forces Reserve USMC - New Orleans, LA

Training & Operations Supervisor/Intelligence Analyst

March 2012 - February 2015

Worked closely with senior leadership to devise organizational requirements for a training and operations reporting system. Developed custom software tools, enabling near real-time management of the training and operational readiness. Prepared readiness reporting and operational briefs which were disseminated to senior leadership.

Key Accomplishments

- Analyzed large amounts of data and developed presentations for senior leadership that addressed current and future training operations and readiness levels.
- Developed and tested scripts written in Visual Basic for Applications (VBA) to enable close integration with the Marine Corps training system. Designed and customized a Microsoft Access database to generate reports and dashboards for senior leadership.
- Credited with facilitating team collaboration across 28 geographically dispersed locations through the development of a battalion wide SharePoint page that integrated closely with a Microsoft Access database capable of delivering information to users in near real time.
- Created documentation and training materials for the use of developed systems
- Awarded a Navy and Marine Corps Commendation Medal for accomplishments.

United States Marine Corps Reserve - Chattanooga, TN

Artillery Section Chief

2004 - 2010 - 2011 - 2012

Set objectives and led a team of 15 in performing operational functions in support of Mike Battery 3rd Battalion 14th Marine Regiment 4th Marine Division USMC. Developed and delivered training to junior associates on all equipment. Played a key role in developing and maintaining international relationships. Met the organization's requirements in inspecting critical components of artillery pieces. Held accountability for all activities that took place on the gun section, including safety procedures and safe handling and storage of ammunition. Developed watch schedules and assigned Marines to working parties.

EDUCATION

Artificial Intelligence Nanodegree – February 2018

Udacity

Deep Learning Nanodegree – May 2017
Udacity

Master of Science in Management Science - Business Analytics – May 2017
University of Alabama in Huntsville – Huntsville, AL

Bachelor of Arts in History – May 2016
University of Alabama in Huntsville – Huntsville, AL

TECHNICAL SKILLS

Nix ■ Python ■ Julia ■ Apache Airflow ■ VueJS ■ Gitlab CI/CD ■ Flask ■ MLFlow ■ Tensorflow ■ Neo4j ■ Shell
Scripting ■ Lua ■ JavaScript ■ Linux ■ MongoDB ■ Ansible ■ Dask ■ Docker ■ Kubernetes